

YACC PROGRAMS

**Assignment No. 4 : Write a program to
evaluate an arithmetic expression, built-in
functions, and variables using YACC
specification.**

Assignment No. 4

4. Write a program

4.A To evaluate an arithmetic expression using YACC tool

4.B To evaluate or check built-in functions using YACC tool

4.C To recognize valid variable name using YACC tool.

Assignment No. 4

4.A To evaluate an arithmetic expression using YACC tool

Input : $5 + 8 * 2$

Output : 21

4.B To evaluate or check built-in functions using YACC tool

Input : `sqrt(25)` [Hint : include `math.h`]

Output : 5

Input : `strlen("India")` [Hint : include `string.h`]

Output : 5

4.C To recognize valid variable name using YACC tool.

Input : `Stud_Name1`

Output : valid variable

4.C To recognize a valid variable name

Lex file

- %{

```
#include "y.tab.h"
```

```
#include<stdio.h>
```

```
#include<stdlib.h>
```

```
%}
```

```
%%
```

```
[a-zA-Z]    return A;
```

```
[0-9]       return N;
```

```
" _"       return U;
```

```
\n         return 0;
```

```
.         return yytext[0];
```

```
%%
```

4.C To recognize a valid variable name

YACC file

```
%{  
    #include<stdio.h>  
    #include<stdlib.h>  
%}  
%token A N U  
%%  
a : AN  
   |aA  
   |aN  
   |aUa  
   |A  
   ;  
%%
```

```
int main()  
{  
    printf("enter the string");  
    yyparse();  
    printf("valid variable");  
}  
void yyerror()  
{  
    printf("invalid variable");  
    exit(0);  
}
```